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Possible origins of Permian remagnetizations in Devonian and Carboniferous limestones from the Moroccan Anti-Atlas (Tafilalet) and Meseta

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Permian overprints have been recognized for several years in different regions of the Hercynian chain. This process seems to affect not only large internal parts, but also external zones of the chain. Devonian limestones sampled in the Tafilalet and Maider massifs near Erfoud and Msissi reveal a strong Permian component carried mainly by magnetite. Scanning Electron Microscope (SEM) and X-ray investigations were performed in order to unravel the origin of this magnetite. Permian remagnetizations are also present in Visean limestones from the Meseta. Isothermal Remanent Magnetization studies and SEM analysis of magnetic separates show that this magnetization is carried mainly by pyrrhotite which could be a result of oxidation of pre-existing pyrite.

1. Introduction

In 1984, a large sampling program was carried out in Morocco. This program principally concerned the Palaeozoic series of the Tafilalet and Maider massifs near Erfoud and Msissi (Fig. 1) where a sedimentary sequence of Lower Devonian to Lower Carboniferous age is well exposed. A total of ten sites were cored into Mid-Devonian to Upper Devonian limestones, four near Erfoud and six around the village of Msissi. The second part of the sampling was done in the Moroccan Meseta south of Rabat, near Azrou and Oulmes (Fig. 1). Thirteen sites were sampled, eight south of Rabat in Visean/Upper Carboniferous limestones, three near Oulmes in Visean sandy limestones, and two near Azrou in Visean/Namurian siliceous limestones. The main purpose of this study was to complete the poorly documented polar wander-path for Africa in Late Paleozoic times.

2. Paleomagnetism

2.1. Techniques

The samples were cored using a portable gasoline-powered drilling machine, and carefully orientated with both a magnetic or sun compass and an inclinometer. One site consists of 10–20 samples which were distributed as widely as possible over the outcrop, generally separated by a few metres. The cores were cut in the laboratory into 2.5 cm diameter and 2.2 cm length specimens. Remanent magnetization was measured using a fluxgate 'DIGICO' magnetometer with a noise level of 3×10^{-5} A m^{-1}. The specimens were treated by stepwise thermal demagnetization re-

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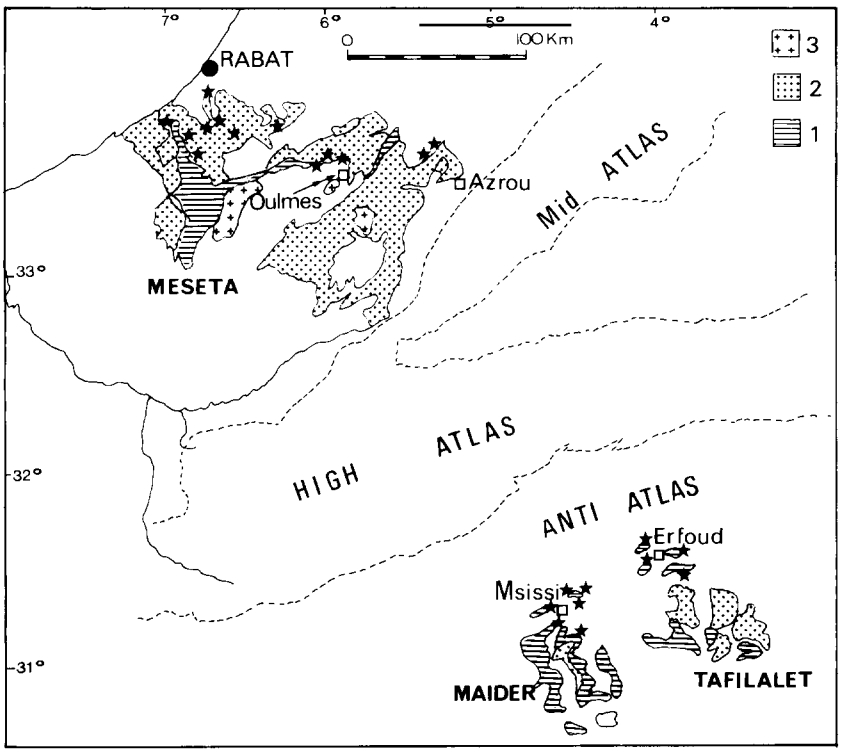


Fig. 1. Sketch map with sampling localities (★) 1, Devonian; 2, Carboniferous; 3, Hercynian granites.

TABLE I

Site mean directions for the Visean Meseta sites before tilt correction and regional mean directions before and after tilt correction

Site no.	n	D	I	k	α_{95}
10	9	144.5	3.1	70.2	6.1
34	3	132.2	6.8	137.6	10.5
35	3	144.2	2.5	64.1	8.4
36	6	150.7	2.2	802	8.8

Means:

$D_m = 142.9$, $I_m = 3.7$, $N = 4$, $k = 102.8$, $\alpha_{95} = 9.1$ before tilt correction.

$D_m = 144$, $I_m = -4$, $N = 4$, $k = 10.8$, $\alpha_{95} = 28$ after tilt correction.

n, Number of specimens; D, declination; I, inclination; k, precision parameter; α_{95}, radius of the circle of confidence; N, number of sites.

aching temperatures of 600 °C or higher. The direction remaining after each demagnetization step was plotted on Zijderveld (1967) orthogonal vec-

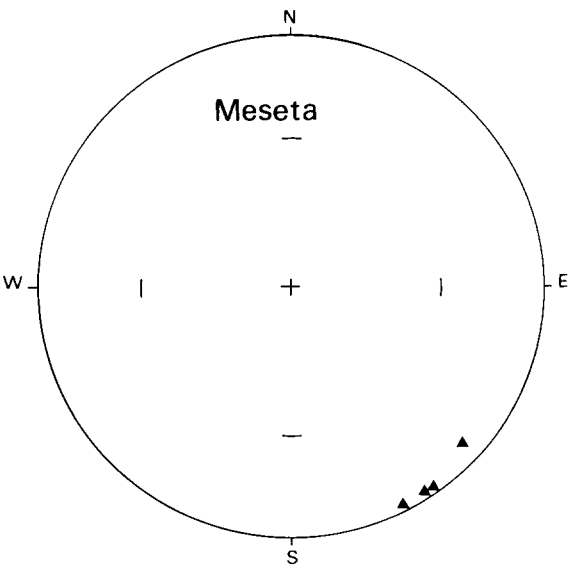


Fig. 2. Characteristic mean directions from the Meseta before tectonic correction.

tor projections and treated by a least squares method.

2.2. Results

2.2.1. Meseta

Four sites (three near Oulmes and one near Azrou, Fig. 1) have mean directions with southeasterly declinations around $D = 140°$ and shallow inclinations (Table I). Folding is important in this area and the major deformation episode is characterized by the Oulmes granitic intrusion, which is considered as synkinematic by Diot et al. (1987). The crystallization age has been dated at 298 Ma by Mrini (1985) (Stephanian stage according to Odin and Gale (1982)) using Rb–Sr methods and is confirmed by the metamorphism of host mica schists yielding a K–Ar age of 290 ± 6 Ma (Huon, 1985). The fold test in the limestones is negative at 99% (Table I) according to McElhinny (1964). These directions are considered as post-folding. Therefore, only directions before tectonic correction have been plotted on Fig. 2. The comparison with African data (Fig. 4) suggests a Permian age for this component.

TABLE II

Paleomagnetic data for the Anti-Atlas

Site no.	n	D	I	k	α_{95}
Ta 1	22	141.8	3.3	67.9	3.7
Ta 2	7	136.1	1.3	93.3	6.2
Ta 3	13	142.8	−0.4	137.5	3.5
Ta 4	12	138.6	−3.2	60.2	5.5
Ta 5	7	138.9	−10.7	33.5	10.4
Ta 6	11	132.1	5	58	5.9
Ta 7	8	142.7	−3.4	102	5.4
Ta 8	10	137.8	−15	78.3	5.4
Ta 10	4	138.2	6.6	58.4	12

Means:

$D_m = 138.8$, $I_m = -1.8$, $N = 9$, $k = 104.7$, $\alpha_{95} = 5$ before tilt correction.

$D_m = 138$, $I_m = -2.7$, $N = 9$, $k = 14.4$, $\alpha_{95} = 13.6$ after tilt correction.

Same legend as for Table I.

2.2.2. Anti-Atlas

In the Anti-Atlas, except for a low-temperature component (160–240 °C) in the vicinity of the present-day field, the samples of the sites located near Erfoud (Fig. 1) exhibit a single stable component up to 580 °C (Fig. 9a). For the sites situated near Msissi, unblocking temperatures of 350–400 °C can be observed. Above this tempera-

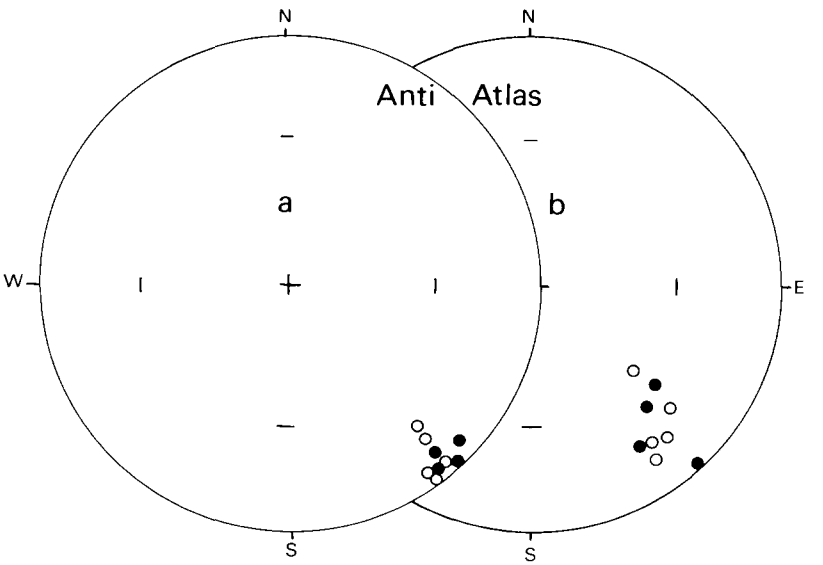


Fig. 3. Mean directions from the Anti-Atlas (a) before tilt correction, (b) after tilt correction. The directions are given in Table I. ○, negative inclinations; ●, positive inclinations.

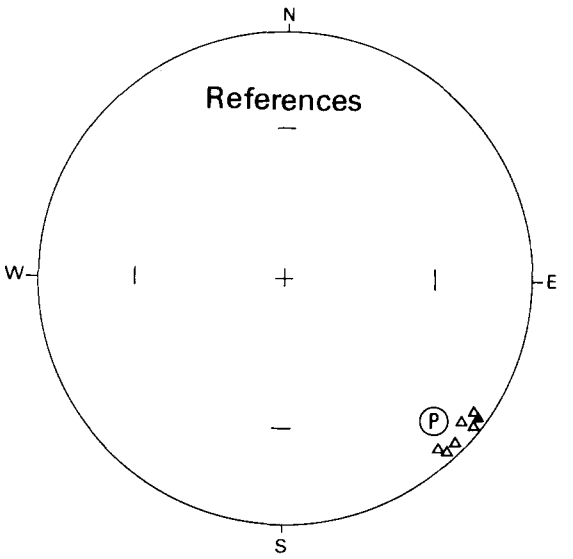


Fig. 4. Reliable Permian directions known for Africa, recalculated in Erfoud geographic co-ordinates from Martin et al. (1978), Hamzeh (1973), Morel et al. (1981), and Daly and Pozzi (1976). △, negative inclinations; ▲, positive inclinations.

ture the intensity of magnetization is too low to allow the determination of another component. In the two cases, the isolated directions are similar, with south-easterly declinations around $D = 140°$ and shallow inclinations, $I = -10°$. On the stereographic projection of Fig. 3, the mean directions of each site are plotted before and after tilt correction (see also Table II). All these sites were sampled in the external part of the Hercynian chain, in an area where folding is weak. The series was tilted by about 20–30° during the Late Devonian (Frasnian/Famennian boundary) as described by Wendt (1985). Before tilt correction, the mean direction is $D_m = 139°$, $I_m = -1.8°$ for nine sites. After tilt correction the mean direction does not change much, $D_m = 138°$, $I_m = -2.7°$, but k decreases from 104.7 to 14.4 whereas α_{95} increases from 5° to 13.6°. The fold test is therefore negative and significant at the 99% level of confidence. The comparison with reliable directions of Africa (Fig. 4) shows that these directions are in very good agreement with Permian data.

Obviously, a widespread Permian overprint has affected the Moroccan (Meseta) Hercynian chain as well as the external parts (Anti-Atlas).

For several years, Permian overprints have been recognized in different regions of the Variscan belt —Armorican massif (Edel and Coulon, 1984; Perroud et al., 1984), Ardennes (Edel and Coulon, 1987), Montagne Noire (Courtillot et al., 1986); and in the external zones—Mauritania (Perrin, 1987), Sahara (Aifa et al., 1987) and Brabant (Edel and Coulon, 1987). This problem is crucial for paleomagnetic interpretation, because if in the future the remagnetization mechanisms are completely understood, it will perhaps be possible to choose outcrops which have not suffered from overprints and are therefore suitable for the determination of reliable Early Paleozoic poles.

3. Rock magnetism

Rock magnetic experiments were done in order to determine the different carriers of this Permian overprint.

3.1. Meseta

In sites where the Permian overprint is well expressed, except for the low-temperature component close to the present-day field, the Zijderveld orthogonal plots are univectorial from 160 to 450 °C and trend towards the origin (Fig. 5a). At higher temperatures, the intensity is too low and the directions can no longer be resolved.

According to rock magnetic experiments, pyrrhotite and magnetite seem to carry this magnetization. Several observations support this idea.

(1) More than 80% of the magnetization is destroyed below 350–400 °C (Fig. 5b).

(2) Isothermal remanent magnetization (IRM) acquisition curves before heating (Fig. 5c) reveal the presence of a low-coercivity mineral. After heating at 500 °C the IRM intensity increases drastically owing to mineralogical changes which form a more low-coercivity material, probably magnetite.

(3) IRM demagnetization curves before heating (Fig. 5d) indicate the presence of two Curie points at 350–400 °C and at 580 °C.

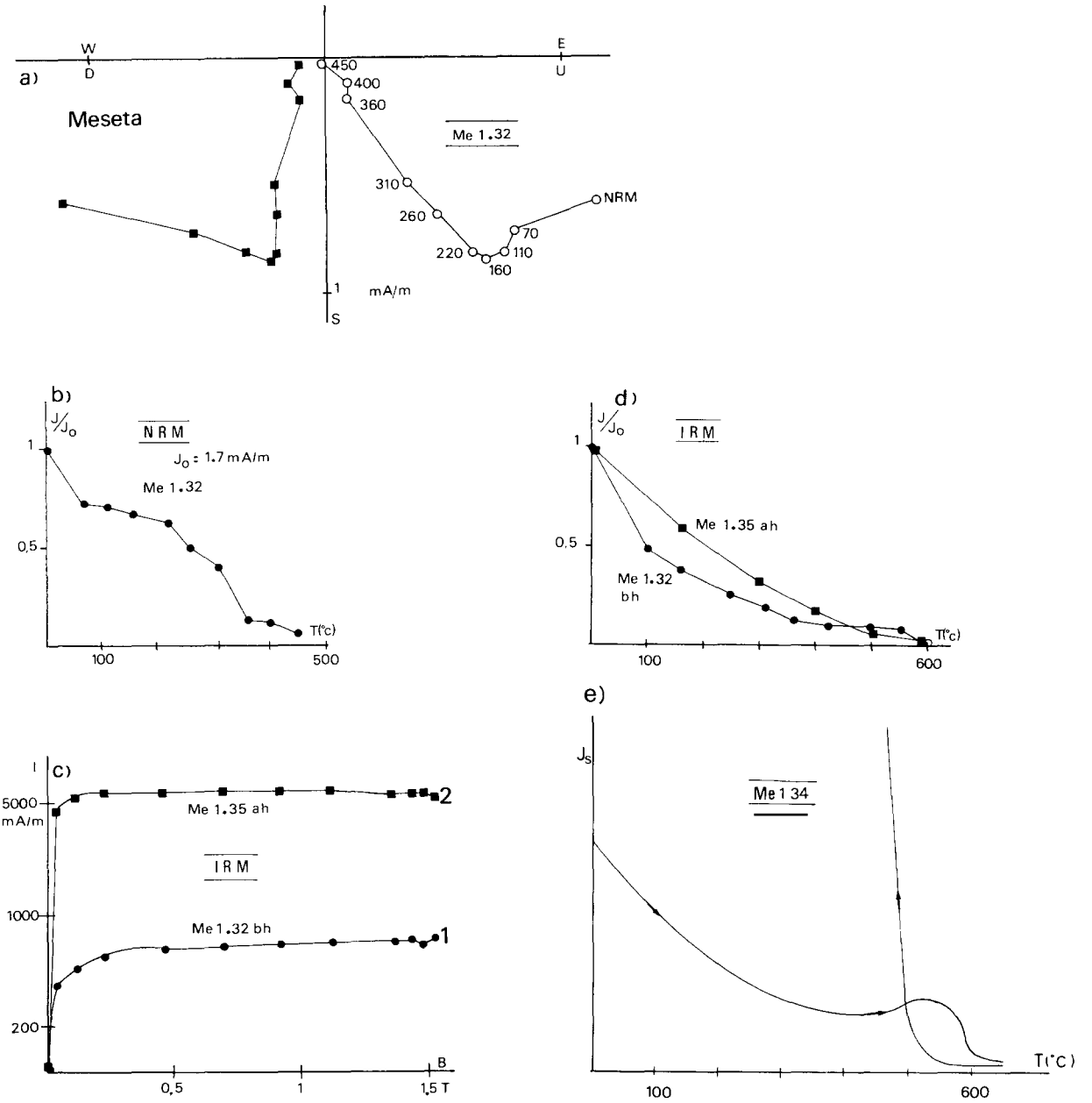


Fig. 5. (a) Zijderveld orthogonal vector projection on the horizontal plane (○) and on the vertical E–W plane (■) for a specimen from Meseta. (b) NRM thermal demagnetization curve. (c) IRM acquisition curve (1) before heating and (2) after heating at 500 °C. IRM demagnetization curves (●) before heating and (■) after heating. (e) $J_S(T)$.

(4) High field magnetization shows an irreversible mineralogical transformation during heating above 400 °C (Fig. 5e).

(5) IRM acquisition in low fields shows a rapid intensity increase.

All these points show that the natural remanent magnetization is carried by a low-coercivity mineral, with maximum unblocking temperatures in the range 350–400 °C. Among the different possibilities we can eliminate goethite, which has a

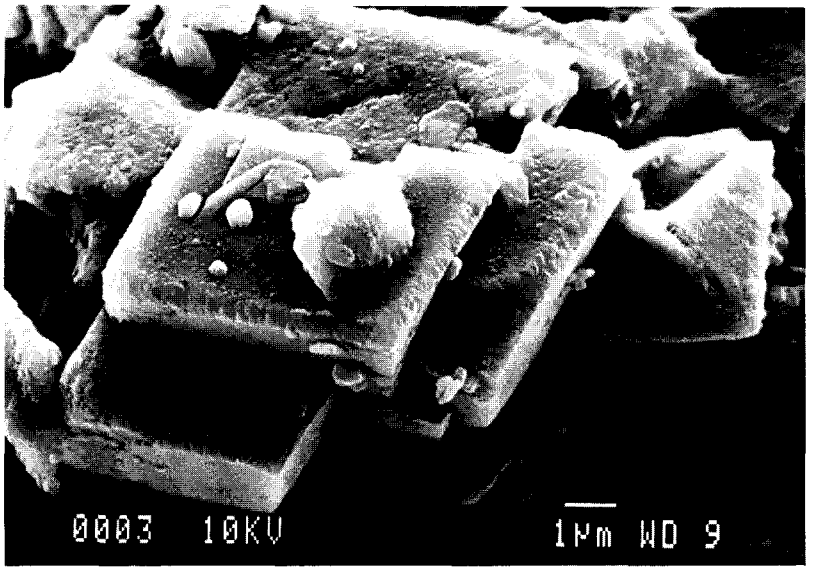


Fig. 6. SEM photomicrograph of hexagonal pyrrhotite from the Meseta. Mean analysis values: $SO_2 = 54.3\%$, $Fe_2O_3 = 45.7\%$.

Curie point about 100–150 °C. Maghemite is unstable around 350–400 °C, but is usually transformed to haematite. The saturation IRM intensity increase is inconsistent with this kind of transformation.

Thus the only mineral possible is pyrrhotite, a sulphide that transforms into magnetite during heating. IRM curves of sulphides and magnetite are similar above 0.1 T, but for sulphides the intensity increases only slowly below fields of $\geqslant$ 0.07 T (Freeman, 1983). As the intensity of IRM increases rapidly below 0.05 T, magnetite seems to be present in these limestones. This observation is consistent with the 580 °C Curie point observed in

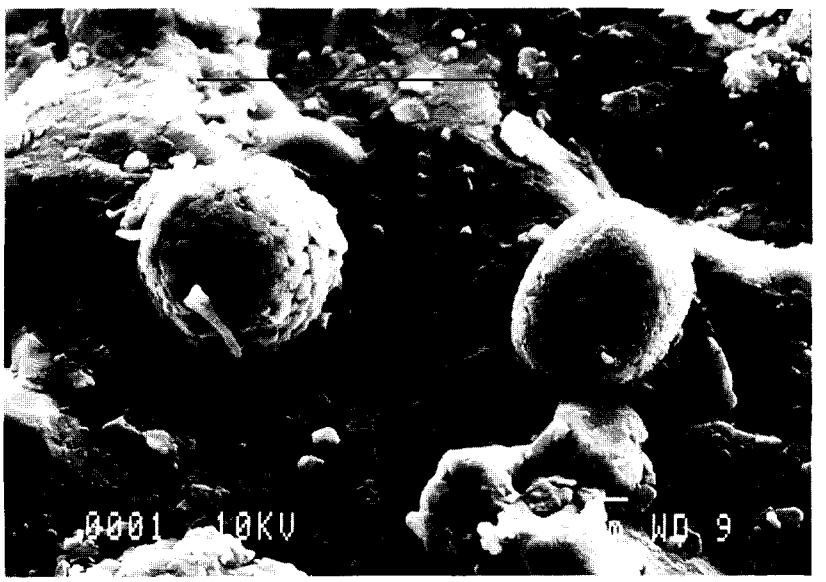


Fig. 7. SEM photomicrograph of framboidal minerals from the Meseta which contain Fe and S as the only elements.

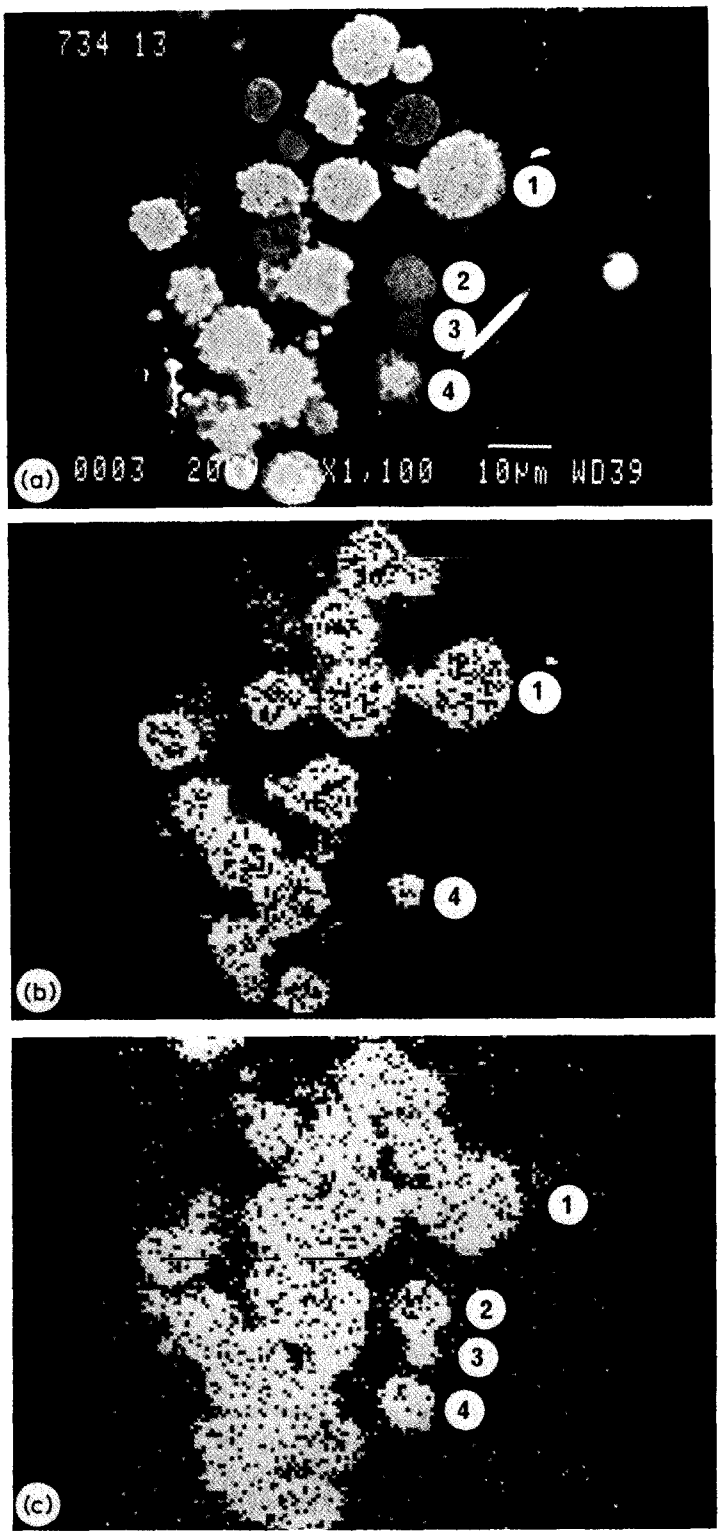


Fig. 8. Framboidal minerals observed with an SEM (thin sections). (a) Analysis points: (1) 42% Fe_2O_3, 57.8% SO_2, (2) 92% Fe_2O_3, 1% SO_2, (3) 92% Fe_2O_3, 1% SO_2, (4) 46% Fe_2O_3, 53% SO_2. (b) Repetition map for S (concentration increases with the white colour). (c) Repetition map for Fe (concentration increases with the white colour, no distinction between Fe^{2+} and Fe^{3+}).

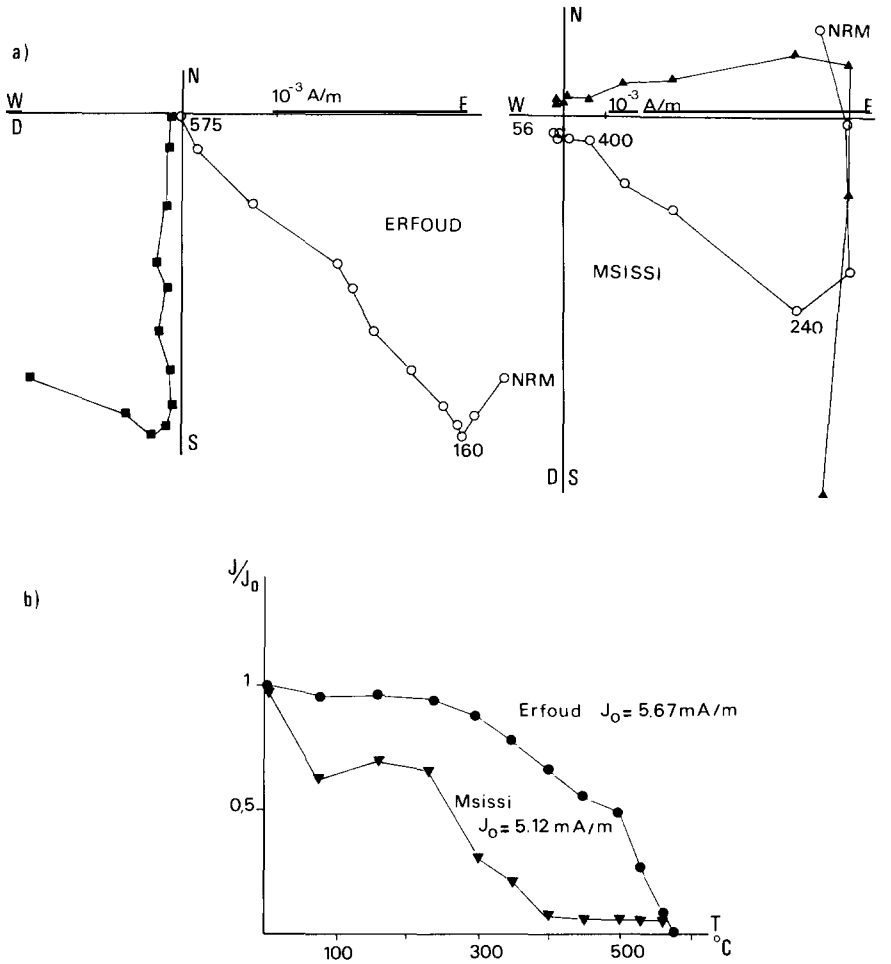


Fig. 9. Devonian limestones from Erfoud and Msissi. (a) Orthogonal vector projection, on the horizontal plane (circles) and on the vertical east–west (squares) or north–south (triangles) plane. (b) NRM demagnetization curve; J_0 initial intensity, J, intensity after heating and cooling at each step of temperature.

Fig. 5d. All these points lead us to conclude that the major part of the remanence is carried by pyrrhotite, but some magnetite fraction probably exists in our samples.

In order to confirm this idea, the magnetic extraction procedure described by Freeman (1986) was applied: after pulverization of the samples and dissolution of the carbonate matrix with acetic acid, the slurry was sent through a tube in a Franz electromagnet. The magnetic particles were separated and put onto a double-face adhesive tape placed on a SEM sample-holder. The minerals were then observed with the SEM (JEOL JSM 40) of the Geology Institute at Strasbourg, which is coupled with a TRACOR allowing qualitative X-ray microanalysis (equivalent to an EDAX system).

Sulphides were observed in several samples. The minerals of Fig. 6 contain Fe and S as the only elements; the habit is that of hexagonal pyrrhotite. In the same sample, a second mineral which contains Fe and minor S, looking like framboidal pyrite, was observed (Fig. 7). The shape of this mineral is similar to that of magnetite interpreted as diagenetic after a sulphide precursor by McCabe et al. (1983) and Wisniowiecki

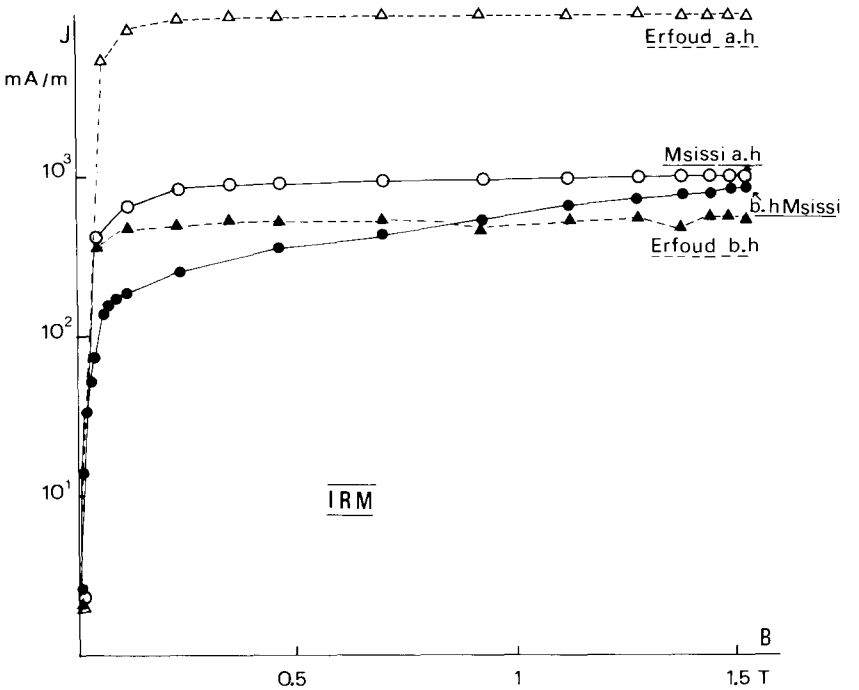


Fig. 10. IRM acquisition curve for a specimen from Anti-Atlas near Erfoud (triangles) before (b.h) and after heating (a.h) at 580 °C, and for a specimen from the Anti-Atlas near Msissi (circles) before and after heating at 590 °C.

et al. (1983). In another sample, these framboidal minerals were observed (in thin sections) with Fe as a single element (Fig. 8) or with a ferroan-sulphide core and a ferroan crust. Thus we assume that this mineral could be diagenetic magnetite which took part in the secondary magnetization.

In conclusion, the major part of the remanence of these limestones (i.e., the Permian overprint) is carried by pyrrhotite, but magnetite also exists in these rocks. The origin of pyrrhotite is probably diagenetic with the primary mineral being pyrite; a slow oxidation can then produce magnetite, as pointed out by Schwartz (1975). This kind of transformation has also been invoked in the case of Spitsbergen limestones by Jelenska and Vincenz (1987). For Mesetian limestones, we conclude that the Permian remagnetization is due to pyrite oxidation. Oxidizing conditions (probably occurring during Permian times) could be due to uplift and contact with an oxidizing atmosphere, or could be due to oxidizing fluid migration related to a major geological event (plate collision), as described by Oliver (1986). However, we cannot exclude completely a detrital origin of the magnetite.

3.2. *Anti-Atlas*

In the Devonian limestones from Erfoud, there is no evidence of sulphides. The orthogonal vector plots indicate the presence of a stable component which trends towards the origin from 160 to 580 °C (Fig. 9a). IRM acquisition curves are typical of a low-coercivity mineral such as magnetite (Fig. 10). The J/J_0 versus temperature curve indicates the presence of a 580 °C Curie point (Fig. 9b). Therefore it seems probable that the magnetization is carried by pure magnetite. It was impossible to observe any magnetite mineral under the SEM; the only ferroan mineral observed is a cubic mineral (Fig. 11). The shape of this mineral strongly suggests a pseudomorphic origin from pyrite. It probably consists of haematite rather

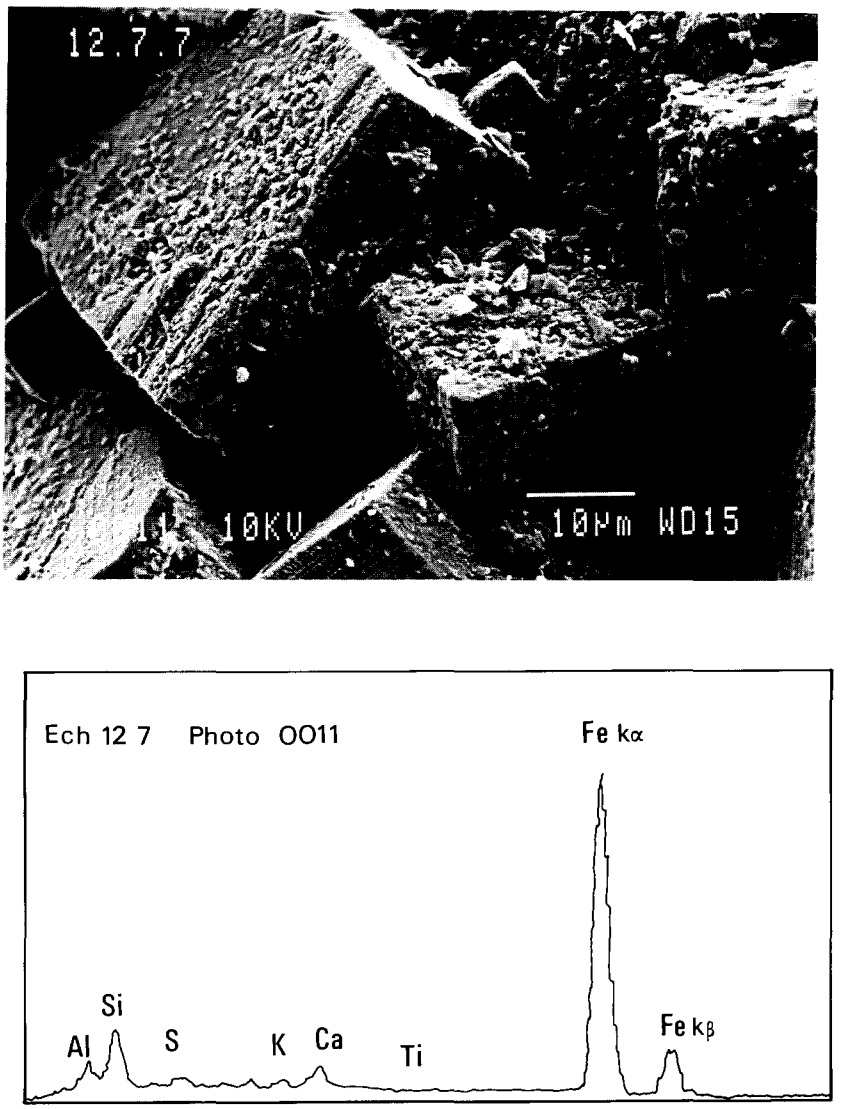


Fig. 11. SEM photomicrograph of cubic minerals from the Anti-Atlas near Erfoud which contain only Fe. The habit of this mineral suggests a pseudomorphic origin, the precursor being pyrite. The new mineral is probably haematite. The lower figure shows the EDAX spectrum obtained from this mineral.

than magnetite and reveals the occurrence of oxidation phenomena in these rocks. This observation is inconsistent with the rock magnetic data. Therefore we have to consider the possibility that our extracts are not representative of the magnetic mineralogy or that the size of the magnetite grains is below the resolution limit of the SEM.

In Msissi Devonian limestones, the demagnetization curves are linear from 240 °C up to 400–450 °C (Fig. 9a). At this temperature most of the remanence is destroyed, suggesting the presence of pyrrhotite, maghemite or titanomagnetite (Fig. 9b). At higher temperature, a low-intensity component persists which is difficult to remove completely and is probably carried by magnetite or haematite. IRM acquisition curves before heating (Fig. 10) are typical of a low-coercivity mineral, which could be magnetite (with Ti or not), pyr-

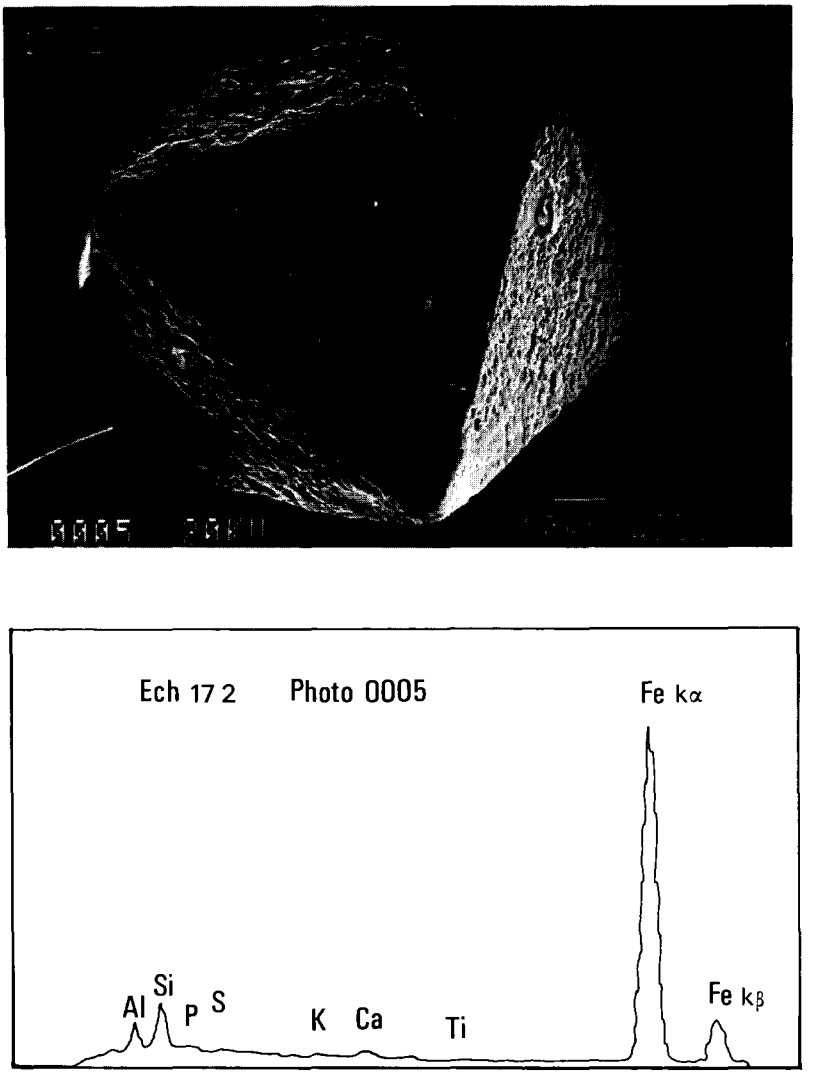


Fig. 12. SEM photomicrograph of an octahedral mineral containing Fe, probably detrital magnetite from the Anti-Atlas near Msissi. The lower figure shows the EDAX spectrum obtained from this mineral.

rhotite or maghemite, with a small high-coercivity mineral contribution (such as haematite). After heating at 590 °C, the intensity of IRM increases only slightly and the acquisition curve indicates only the occurrence of a low-coercivity mineral. It is difficult to determine precisely the carriers of the remanence: pyrrhotite converts to magnetite, and this transformation seems to produce an important intensity increase, which was not observed; also, the unblocking temperature range is a little higher for pyrrhotite. Maghemite converts to haematite, but this process is not evident in our samples. Therefore the possibility exists that titanomagnetite is the dominant mineral, but the other possibilities cannot be excluded completely.

Some typical ferroan octahedral minerals (Fig. 12) which look like detrital magnetite were observed with the SEM, but no sulphide phases seem to exist in the extracts. A detrital origin of the magnetite is inconsistent with the Permian overprinting and with the 400 °C 'Curie point' (this mineral does not contain Ti); we have also to

envisage a pseudomorphic transformation due to oxidation of the detrital magnetite into another mineral, or that these detrital minerals do not contribute to the remanence. The same conclusion drawn at Erfoud should be envisaged. The detrital minerals are probably transformed and do not take part in the NRM. Therefore another magnetic phase, which seems to be diagenetic magnetite as suggested by rock magnetism, should be present which cannot be observed at this time. This hypothesis needs to be confirmed by further X-ray or microprobe analysis to resolve the exact chemical composition of the minerals.

4. Conclusion

Permian overprinting is omnipresent in the Hercynian chain. Its origin is one of the great problems in paleomagnetism. Probably different mechanisms exist, because, in the Meseta, pyrrhotite and magnetite carry the Permian component, and, in the Anti-Atlas, diagenetic magnetite (despite the fact that it was not observed) seems to be responsible for the overprint magnetization. Therefore different processes have to be accounted for in order to explain this observation. We notice that the Permian overprint has a larger extent in the external domains of the Hercynides. In the Meseta, some sites reveal the occurrence of older magnetization components which will be described elsewhere, but this is not true in the Anti-Atlas. This observation could reveal that the main overprinting mechanism is diagenetic because in the external zones metamorphism and tectonism (the other principal cause of remagnetization) are of less importance. A thermoviscous event as suggested by Kent (1985) seems to be irrelevant. The origin of the magnetite is not clearly documented and needs further investigations especially in the Anti-Atlas. In the Meseta, we propose one remagnetization mechanism due to the presence of sulphides; their instability during oxidation seems to play an important role in these overprint problems. This oxidation could be related to fluid migration or uplift and contact with the oxidizing atmosphere. It is evident that in other regions where the Permian component is present, sedimentology and environmental conditions were quite different, as well as the age of the rocks considered. Therefore sulphide oxidation cannot always be envisaged and in the Meseta this mechanism is certainly not the only one occurring.

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References

Aifa, T., Feinberg, H. and Pozzi, J.P., 1987. A new paleopole for Africa at the Devonian/Carboniferous boundary: implication for Hercynian geodynamics. Terra Cognita, 7: 177.

Courtillot, V., Chambon, P., Brun, J.P. and Rochette, P., 1986. A magnetotectonic study of the Hercynian Montagne Noire (France). Tectonics, 5: 733–751.

Daly, L. and Pozzi, J.P., 1976. Résultats paléomagnétiques du Permien inf. et du Trias Marocain, comparaison avec les données africaines et Sud-américaines. Earth Planet. Sci. Lett., 29: 71–80.

Diot, H. Bouchez, J.L., Boutaleb, M. and Macaudière, J., 1987. Le granite d'Oulmès (Maroc Central): structure de l'état magmatique à l'état solide et modèle de mise en place. Bull. Soc. Géol. France, 111, 1: 157–168.

Edel, J.B. and Coulon, M., 1984. Late Hercynian remagnetization of Tournaisian series from the Laval syncline, Armorican massif, France. Earth Planet. Sci. Lett., 68: 343–350.

Edel, J.B. and Coulon, M., 1987. A paleomagnetic cross-section through the Ardenne and the Brabant massifs. J. Geophys., 61: 21–29.

Freeman, R., 1983. The magnetization of some pelagic Tethyan limestones. Dissertation, ETH Zurich, 168 pp.

Freeman, R., 1986. Magnetic mineralogy of pelagic limestones. Geophys. J. R. Astron. Soc., 85: 433–452.

Hamzeh, R., 1973. Etude Paléomagnétique de Formations Triasiques et Liasiques du Maroc et leurs Implications pour l'Ouverture de l'Atlantique. Thèse 3è cycle, Univ. de Strasbourg, 160 pp.

Huon, S., 1985. Clivage Ardoisier et Réhomogénéisation Isotopique K–Ar dans des Schistes Paléozoïques du Maroc. Thèse, Univ. de Strasbourg, 124 pp.

Jelenska, M. and Vincenz, S.A., 1987. Origin of the magnetization of Permo-Carboniferous sediments of Spitzbergen, Svalbard archipelago. Earth Planet. Sci. Lett., 85: 173–182.

Kent, D.V., 1985. Thermoviscous remagnetization in some Appalachian limestones. Geophys. Res. Lett., 12: 805–808.

Martin, D.L., Nairn, A.E.M., Noltimier, H.J., Petty, M.H. and Schmitt, T.J., 1978. Paleozoic and Mesozoic paleomagnetic results from Morocco. Tectonophysics, 44: 91–114.

McCabe, C., Van der Voo, R., Peacor, D.R., Scotese, C.R. and Freeman, R., 1983. Diagenetic magnetite carries ancient yet secondary remanence in some Paleozoic sedimentary carbonates. Geology, 11: 221–223.

McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophys. J. R. Astron. Soc., 8: 338–340.

Morel, P., Irving, E., Daly, L. and Moussine-Pouchkine, A., 1981. Paleomagnetic results from Permian rocks of the northern Saharan craton and motion of the Moroccan Meseta and Pangea. Earth Planet. Sci. Lett., 55: 67–74.

Mrini, Z., 1985. Age et Origine des Granitoïdes Hercyniens du Maroc: Apport de la Géochronologie et de la Géochimie Isotopique. Thèse, Univ. de Clermont-Ferrand, 150 pp.

Odin, G.S. and Gale, N.H., 1982. Mise à jour de l'échelle des temps calédoniens et hercyniens. C.R. Acad. Sci., Ser. D, 294: 453–456.

Oliver, J., 1986. Fluids expelled tectonically from orogenic belts: their role in hydrocarbon migration and other geologic phenomena. Geology, 14: 99–102.

Perrin, M., 1987. Paléomagnétisme des Séries Rouges à Aimantations Multiples d'Âge Protérozoïque Paléozoïque d'Afrique et d'Amérique du Nord. Thèse, Univ. de Montpellier, 365 pp.

Perroud, H., Robardet, M., Van der Voo, R., Bonhommet, N. and Paris, F., 1984. Revision of the age of magnetization of the Montmartin red beds, Normandy, France. Geophys. J. R. Astron. Soc., 80: 541–549.

Schwartz, E.J., 1975. Magnetic properties of pyrrhotite and their use in applied geology and geophysics. Geol. Surv. Canada Paper, 74, 59: 1–24.

Wendt, J., 1985. Disintegration of the continental margin of northwestern Gondwana: Late Devonian of the Eastern Anti-Atlas. Geology, 13: 815–818.

Wisniowiecki, M., Van der Voo, R., McCabe, C. and Kelly, W.C., 1983. A Pennsylvanian paleomagnetic pole from the mineralized Late Cambrian Bonneterre formation, southeast Missouri. J. Geophys. Res., 88: 6540–6548.

Zijderveld, J.D.A., 1967. A.C. Demagnetization of rocks. Methods in palaeomagnetism. In: K.M. Creer, S.K. Runcorn and D.W. Collinson (Editors), Developments in Solid Earth Geophysics, Vol. 3. Elsevier, Amsterdam, 609 pp.